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Breast Cancer in Singapore: Trends in Incidence 1968–1992

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Seow A (Department of Community, Occupational and Family Medicine, National University of Singapore, National University Hospital, Lower Kent Ridge Road, Singapore (0511)), Duffy S W, McGee M A, Lee J and Lee H P. Breast cancer in Singapore: Trends in incidence 1968–1992. *International Journal of Epidemiology* 1996; **25**: 40–45.

Background. Breast cancer is the most commonly occurring cancer among women in Singapore, a country which has experienced significant changes in lifestyle over the past three decades. The increase in incidence of the disease is a matter of some concern.

Methods. Data from the population-based Singapore Cancer Registry for 1968–1992 were used to determine time trends, inter-ethnic differences and the contributions of age, period and cohort effects to the incidence of the disease.

Results. Our results revealed an average annual increase of 3.6% over the 25-year period for all women, from 20.2 per 100 000 women in the period 1968–1972 to 38.8 per 100 000 in 1988–1992. There was a statistically significant difference between the three major ethnic groups, the rate of increase being highest in Malays (4.4%) and lowest in Indians (1.4%). The overall increase was attributable to a strong cohort effect that remained significant when adjusted for time period for Chinese women and for all ethnic groups combined. The risk was observed to increase in successive birth cohorts from the 1890s to the 1960s.

Conclusions. Our results suggest that breast cancer incidence rates are likely to continue to increase more sharply in the future as women born after the mid-20th century reach the high-risk age groups. They also suggest the pattern by which important aetiological factors for the disease in our population have exerted their effects, and provide support for the role of demographic and lifestyle changes as possible risk factors.

Keywords: population-based, age-period-cohort, ethnic groups, incidence trends, breast cancer

The incidence of female breast cancer (ICD-9 174, ICD-0 C509) varies markedly between countries and is highest in the US and Northern Europe, intermediate in Southern and Eastern Europe, and lower in Asia and the Far East.[1] Singapore, with an ethnic composition of 78% Chinese, 14% Malays and 7% Indians, has rates among the highest in Asia (Table 1). Over the past 40 years, changes in lifestyle and environment associated with her transition from a developing to a newly-industrialized economy have pushed the pattern of disease towards that of the West and increased the importance of chronic degenerative diseases as causes of morbidity and mortality in the population.[2]

Data from the Singapore Cancer Registry, which began collecting basic information on all incident cancers from 1968, show a steady increase in incidence rates of breast cancer.[3–5] A review of trends in cancer incidence among Singapore Chinese[6] from 1968 to 1982 noted a significant rise in breast cancer with an age, and possibly a birth cohort effect. This cohort effect, in which women sharing the same period of birth exhibit a common level of risk that reflects the particular experience of the cohort as it progresses through

TABLE 1 *Breast cancer incidence in selected countries 1983–1987*

Country, Region	Age-adjusted incidence/ 100 000 females
United States, Bay Area, White	104.2
United States, Bay Area, Black	81.6
Canada, British Columbia	76.1
Sweden	62.5
Australia, New South Wales	59.6
England and Wales	56.1
German Democratic Republic	46.3
Poland, Warsaw City	38.7
Czechoslovakia	34.5
Hong Kong	32.3
Singapore	31.1
Japan, Miyagi	27.8
India, Bombay	24.6
China, Shanghai	21.2

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life, is of interest because it may suggest environmental or behavioural influences that give clues to breast cancer aetiology. In this paper, we seek to update the analysis of breast cancer trends and to re-examine the contributions of age, period and cohort effects to this trend. We also describe the inter-ethnic differences in trends and discuss implications of our observations in the light of known risk factors for breast cancer.

MATERIALS AND METHODS

The Singapore Cancer Registry is a population-based registry covering the entire resident population of Singapore. Details of procedures, sources of data and coding practices are documented elsewhere.[3] Population data were derived from the 1970, 1980 and 1990 censuses, with interpolation by age and sex between these years. Where age-standardized rates are quoted, these are standardized to the 'World' population[7] for purposes of international comparison.

Data were analysed by Poisson (log-linear) regression.[8] This method allows for estimation of trends across individual calendar years to obtain average annual percentage changes, significance testing using deviance χ^2 tests, and the estimation of relative risks and their confidence intervals.

The Poisson regression procedure fits a model of the form

$$\log(\text{rate}) = a + b \times (\text{covariate } 1) + c \times (\text{covariate } 2) + \ldots$$

To estimate an age-adjusted average annual percentage change we fitted a model

$$\log(\text{rate}) = a_i + b \times (\text{year})$$

with $i = 1, 2, \ldots, 13$, where there are 13 age groups (20–24, 25–29, 30–34, ..., ≥80), and year is given as 1, 2, ..., 25 (year 0 is 1968, year 1 is 1969 and so on up to 1992). The exponential of the estimate of b is the relative risk per passing of each calendar year. Thus, for example, for Chinese women we estimated b to be 0.03524. The exponential of this is 1.036, an average increase in rates of 3.6% per year, as shown in Table 3. The model allows for interaction tests to compare these trends between groups, for example, to ascertain whether the temporal changes vary by ethnic group.

In addition to estimation of changes in time and inter-ethnic differences, we also analysed temporal changes using age-period-cohort models.[9,10] This is the procedure used to assess which, after adjusting for age, of the following models better describes the data: a difference in incidence between time periods; a difference in incidence between birth cohorts; or both types of difference.

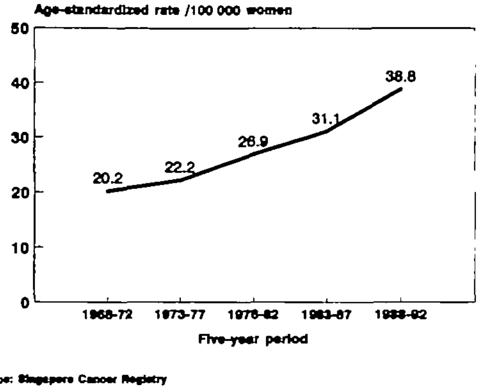

FIGURE 1 *Incidence rates for breast cancer, Singapore 1968–1992*

Information on deaths from breast cancer, and on crude birth rates was obtained from the Registry of Births and Deaths, Singapore for the years 1930 to the present.

RESULTS

Figure 1 shows, on a linear scale, the general trend in incidence for all female residents, which has risen from 20.2 to 38.8 per 100 000 between 1968 and 1992. Table 2 gives details of the period-specific numbers of cases and the age-adjusted rates for each 5-year period, by ethnic group.

In Table 3, the relative risks and average annual change for each major ethnic group are presented. It can be seen that while Malay women have significantly lower risks than their Chinese counterparts, their rate of increase is highest among the three groups. Indian women have an intermediate level of risk, and the lowest rate of increase, although the small number of cases makes interpretation of the overall trend difficult. The changes in rates with time differ significantly between ethnic groups.

Figure 2 shows the changes over time by birth cohort. There is a tendency for successive birth cohorts to have higher age-specific rates compared with previous cohorts, suggesting a strong cohort effect. This is borne out by the results of the analysis (Table 4) which shows for all residents combined a highly significant cohort effect after adjustment for age and period. The period effect did not maintain its significance when adjusted for cohort. A similar pattern was seen when data for Chinese were analysed separately, while for Malays and Indians, neither period nor cohort effects were

TABLE 2 *Number of cases and age-standardized incidence of breast cancer by ethnic group*

Ethnic group	Period				
	1968–1972	1973–1977	1978–1982	1983–1987	1988–1992
Chinese	559	740	1051	1455	2199
	(19.8)	(22.7)	(27.4)	(31.8)	(39.7)
Malay	63	70	111	152	263
	(17.1)	(15.5)	(21.2)	(23.0)	(34.1)
Indian	25	41	52	86	108
	(25.4)	(27.6)	(29.8)	(33.4)	(30.9)
All	676	863	1238	1729	2619
	(20.2)	(22.2)	(26.9)	(31.1)	(38.8)

Figures in parentheses are 'World'-standardized incidence rates/100 000 women/year.

TABLE 3 *Age-adjusted relative risks (RR) and average annual percentage changes in incidence by ethnic group*

Ethnic group	RR	95% CI	Average annual change[a]	95% CI
Chinese	1.00	–	3.6%	3.2%, 4.1%
Malay	0.81	0.74, 0.88	4.4%	3 1%, 5.7%
Indians	0.89	0.79, 1.01	1.4%	–0 1%, 3.1%
All groups			3.6%	3.2%, 4.0%

[a] Significant difference between trends for ethnic groups, $P = 0.02$.

significant when adjusted for each other. The effect of including the 'drift' parameter, which represents a log-linear change in rates not exclusively identifiable as a cohort or period effect, was significant for all groups except Indians.

Table 5 presents the relative risks by cohort for Chinese and Malay women. Among Chinese women, there is a clear increase in risk with succeeding birth cohorts up to those born around 1958. Compared with those born around 1893, these women show a tenfold higher risk of breast cancer. The increase does not appear to continue for the post-1960 cohorts, but it should be noted that there are less data available for these groups. For Malay women, the increase appears to begin from the 1908 cohort, and the risk drops off for the latest cohort in a similar manner. The numbers of cases among Indian women were too few for the risks to be compared meaningfully in this way.

Figure 3 shows the number of deaths from breast cancer recorded in the decade 1980–1990 and the corresponding age-adjusted rates. It is interesting that the mortality rates have remained remarkably constant despite the steep increases in incidence during this period.

DISCUSSION

In summary, our analysis shows that the previously observed increase in incidence of breast cancer has continued and confirms that this is mainly attributable to a strong cohort effect. Between ethnic groups, the overall rate of increase is highest in Malays, and lowest in Indians. The increasing risk with birth cohort up to the 1960s is especially marked among the Chinese population, and remains significant when adjusted for time period.

We believe that the quality of the data used is good. Although notification to the Singapore Cancer Registry is voluntary, the large majority (68.3%) of all cases were notified by medical practitioners, and the remainder on the basis of pathology reports and hospital records (25.8%), with cases registered on the basis of death certificates constituting only 6%. Of breast cancer cases registered, 95.6% had a histological basis for diagnosis.[5]

It is unlikely that changes in coding practices, or diagnostic accuracy over the past 25 years could have given rise to an artificial increase for such a site as breast cancer. As Singapore is a small and densely populated country, it is also unlikely even in the 1960s and 1970s, that cases of breast cancer would have been missed by lack of access to health care, or by failures of the death registration system. The relative lack of mammographic screening in Singapore at present precludes attributing the trend to increased detection of early cancers. The most convincing evidence, however, that the observed trend is real is that it is clearly cohort-related, rather than period-related.

In the interpretation of age-period-cohort analysis, it is important to note the limitations posed by problems of identifiability. The model with all three variables age, period and cohort simultaneously is not fully unique, or identifiable,[9,10] and may be described by different sets

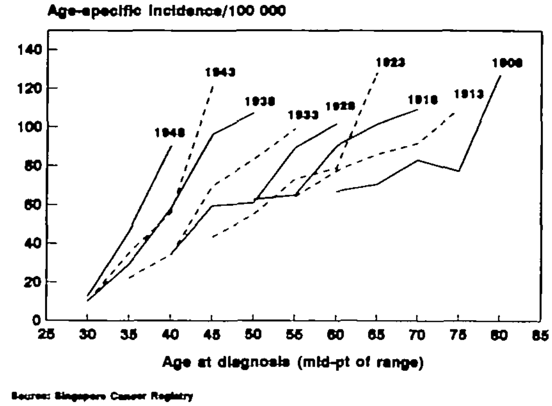

FIGURE 2 *Age-specific incidence of breast cancer by birth cohort, Singapore 1968–1992*

of parameter values. This is illustrated by the fact that a trend test for 'drift' in incidence[10] gives exactly the same result if it is applied to a trend with time period or a trend with birth cohort. When global tests are performed it is possible to choose between the two models, but inference must still be tentative. A model for cohort is often a better fit to the data than a model for period, but the improved fit is achieved at the expense of a considerable increase in the complexity of the model.[9]

The observed increase in incidence, and its relation to period of birth is in accord with findings reported in other parts of the world, including the US, Canada, Europe and China.[11–14] Unlike previous studies which demonstrated a nadir in the cohort effect[15] for women born around 1895 to 1900, we did not find such a trough in relative risk in our data. The observed decrease in risk in cohorts born after 1960 should be interpreted with caution until further data on these groups of women become available in the future.

The strong cohort effect would seem to implicate an aetiological factor that has increased over time while successive cohorts of women are exposed to it during their lifespan. Furthermore, it has been suggested[14] that the parallelism of the age-incidence curves by birth cohort (Figure 2) points to an exposure that exerts its primary effect early in life, before the age of appreciable risk is attained.

The divergence between incidence and mortality rates, with the latter remaining relatively constant despite significant increases in incidence, has been reported in several other studies.[11,13] It is likely that this is related to earlier diagnosis and more effective treatment of tumours. The setting up of a hospital-based breast cancer registry, which is currently being planned, will provide information on the stage at diagnosis and mode of treatment of breast cancer cases, against which these hypotheses may be tested in the future.

Any correlation between trends in known risk factors and the observed trends in breast cancer risk is at the most suggestive. However, our findings do suggest that any factor purporting to significantly influence breast cancer risk among local women must be able to explain the strong cohort effect described. Among the various risk factors for breast cancer that have been investigated in recent years, reproductive factors and some aspects of diet have received acceptance as contributing to differences in rates over time and between countries.[16]

There have been dramatic changes in lifestyle among Singaporeans over the past 50 years brought about by the growing economy and increasing affluence. The greater extent of female participation in the workforce (from 24.6% in 1970 to 51.3% in 1992)[17] and the tendency for women to pursue and establish their careers

TABLE 4 *Female breast cancer in Singapore, 1968–1992. Deviance χ^2 statistics for age, period and cohort effects (factors adjusted for in parentheses)*

Model	χ^2 tests				
	All	Chinese	Malay	Indian	d.f.
Age	5988***	4870***	613***	363***	12
Drift (age)	379***	328***	50***	3	1
Period (age)	384	330	54	5	4
Period (age + cohort)	3	2	5	3	3
Cohort (age + period)	58***	46***	23*	15	14
Cohort (age)	439***	375***	72***	17	15

* $P = 0.05$; *** $P < 0.001$.

TABLE 5 *Relative risks of breast cancer by birth cohort and ethnic group*

Birth cohort[a] (Midpoint)	Chinese		Malays	
	Relative risk	95% CI	Relative risk	95% CI
1893	0.29	0.17–0.52	0.65	0.08–5.42
1898	0.52	0.38–0.69	0.19	0.03–1.49
1903	0.57	0.46–0.71	0.50	0.21–1.17
1908	0.57	0.48–0.68	0.48	0.25–0.93
1913	0.71	0.61–0.82	0.68	0.42–1.09
1918	0.80	0.71–0.91	0.83	0.57–1.23
1923	0.83	0.73–0.93	0.89	0.63–1.26
1928	1.00	–	1.00	–
1933	1.19	1.06–1.33	1.20	0.87–1.64
1938	1.72	1.53–1.94	1.44	1.02–2.05
1943	2.01	1.76–2.29	1.73	1.17–2.57
1948	2.66	2.30–3.07	3.24	2.16–4.87
1953	2.75	2.30–3.29	3.18	1.96–5.15
1958	3.34	2.61–4.27	6.58	3.58–12.11
1963	3.33	2.20–5.04	1.69	0.46–6.24
1968	2.01	0.59–6.91	–	–

[a] Reference period: 1926–1930 (midpoint 1928).

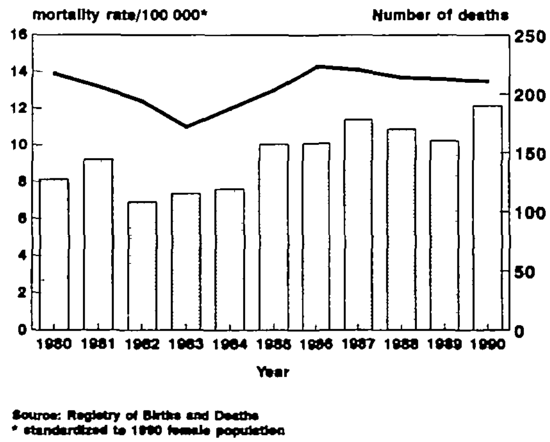

FIGURE 3 *Mortality from breast cancer, Singapore 1980–1990*

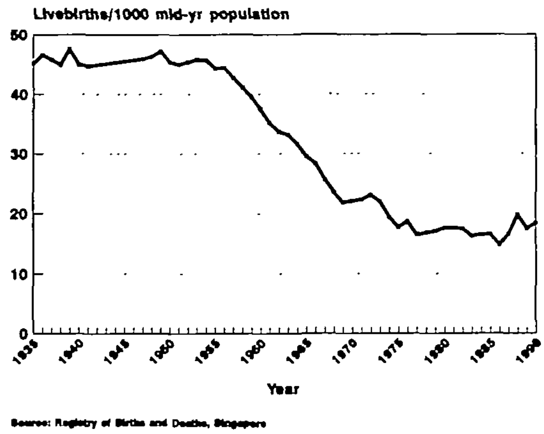

FIGURE 4 *Crude birth rates for Singapore, 1935–1990*

before childbearing has had an impact on fertility, which has been compounded by the successful family planning campaign of the 1970s. This resulted in a fall in birth rates and in particular in early fertility, with the lowest point around 1985, shortly before the new pronatalist policies were introduced (Figure 4). Cohort fertility as a contributor to breast cancer trends was proposed as early as in 1958[18] and examined in the light of more recent trends with varying results.[19,20] Although the available data do not allow us to examine the correlation statistically, the observed trend supports an effect of falling fertility on breast cancer risk. That early fertility and the age at first birth are of importance is in accordance with the parallelism of the age-cohort curves we describe above. What may prove interesting in the future is the effect of the slowing in decline since 1985, as the cohorts entering the reproductive age group at that time correspond to those for whom risk appears to have decreased in our analysis.

Changes in the diet of Singaporeans from the 'traditional' pattern of one's country of origin, to adoption of a more 'Western' diet have been previously described and affect particularly the young.[21,22] Because greater affluence has led to an increase in consumption of almost all food groups,[23] it is difficult to compare trends in consumption of specific items with breast cancer risk at an ecological level. A case-control study conducted in Singapore did find that the main dietary effects were confined to premenopausal women.[24]

Data on other risk factors for breast cancer are not readily available locally, but what evidence there is suggests that age at menarche and attained height also follow the pattern expected as characteristics of behaviour and lifestyle approach those of the West. Two separate studies conducted a decade apart reported a fall in the age at menarche from 14.8 years in 1966[25] to 13.25 years in 1976.[26]

We believe that the existence of a strong cohort effect for breast cancer risk among Singapore women, and in particular among the Chinese, has several important implications. The higher relative risk in cohorts of women who have currently not reached the highest risk age groups implies that if the present trend continues, the overall incidence will continue to increase, and the increase is likely to take place at an even greater rate than at present.

The strong cohort effect, and the fact that the cohorts at highest risk are currently in the premenopausal age group, would account for the increasing numbers of cases being diagnosed in younger women. Up to the present, breast self-examination and clinical breast examination have been the chief means of secondary prevention provided to the population. Mammographic screening, proven effective for women above 50 years of age in many countries, is not widely available, although a project to evaluate its efficacy among Singapore women is about to be implemented. Given the epidemiology of the disease in this country, the appropriate means of prevention for younger women remains an issue. At the same time, the notion that risk is determined even earlier in life suggests that any primary prevention measures that become accepted should be targeted on the appropriate age group.

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